COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-APP-011: Front End Website and Web Application Framework

Document ID	TS-APP-011
Title	Front End Website and Web Application Framework
Executive Sponsor	CTO – Chief Technology Office, Director of Services Engagement
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.0
Effective Date	05/23/2019

1. Purpose

OIT is statutorily mandated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy stakeholder needs surrounding User Interface and User Experience, this document is submitted as the standard for specifications, methodologies, and practices to provide the service.

This document communicates the standards OIT has developed with the intent of providing enterprise standard configurations for a standardized Front End Website and Web App Development Framework, based on a best-of-value decision assessment involving stakeholder needs, hardware configurations, and operational and security best practices.

The ultimate purpose of this standard is to describe the minimum requirements for the Front End Website and Web App Development Framework. These cascading style sheet (CSS) standards ensure a consistent application of style to a project, making maintenance easier and minimizing style conflicts, while meeting minimum requirements for accessibility for government.

2. Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended

as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

CTO - Chief Technology Office - Director of Services Engagement

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Supports, trains, and consults on this technical standard and all accompanying guidance documents and references.
- Presents to the Architecture Review Board for approval and adoption.

CTO - Application Service Managers and Developers

- Implement this standard across the OIT portfolios.

CTO - QA/Testing

- System testers will ensure OIT-approved UX standards are implemented as part of their testing process.
- Usability will be evaluated as part of the User Acceptance Test process.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Goals

- Maintain consistency in visual elements and defining behavior.
- Ensure a seamless, intuitive, easy-to-use digital experience.
- Adhere to a common State of Colorado look and feel when developing a UI.
- Make maintenance easier and minimize style conflicts.

Process

- Any application created by OIT will use the CSS detailed in the State of Colorado Style Library. If the application does not allow use of the Style Library's stylesheet, developers should mimic the UI components as closely as possible using available CSS/styling options, and the application should still meet minimum accessibility requirements (per TS-OEA-001).
- Developers and project teams should use the Style Library as a resource for templates and best practices when building a web front end or validating a user interface built by a third party.

- Application design exceptions should be submitted to the UX Experience Manager for approval.

Requirements

The order of preference for CSS implementation is:

- External CSS file (State of Colorado Style Library): Should always be used if there is any reuse of style.
- Page-level CSS (using the Style Library): Page-embedded styling (in the document HEAD) is appropriate for page-specific, non-reusable style.
- Inline attribute CSS styles (using the Style Library): Inline element style should mostly be avoided, but may be necessary on occasion.
- Add CSS through external files, minimizing the number of files where possible; the stylesheet link should be placed in the HEAD of the document.
- Use the LINK tag to include stylesheets rather than @import, to improve efficiency.
- Avoid inline styling and minimize inline styles in general; separation of structure and style makes maintenance easier.
- Elements that occur only once inside a document should use IDs for style hooks; otherwise, use classes.
- Understand cascading and selector specificity so terse and effective style selectors can be written.
- Write selectors optimized for speed; avoid expensive CSS selectors such as the wildcard selector, and don't qualify ID or class selectors.
- Understanding of the CSS and browser-based box model is necessary for conquering the fundamentals of CSS layouts.
- CSS validators can help identify faulty CSS.
- Single-line CSS should be avoided, as it can cause version control issues and is less readable for multiple authors.
- Alphabetizing CSS properties increases the odds of larger repeatable patterns being present, which aids GZIP compression.
- Use namespaces for custom CSS rules to better distinguish custom styles from those implemented by frameworks or plugins.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the “Owner of Standard” listed under Roles and Responsibilities.

Requests may be acquired by completing the “EXCEPTION REQUEST to Technical Standard” form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

- OIT documents and policies - State of Colorado Style Library; TS-OEA-001: Technology Accessibility for Persons with Disabilities
- State documents and policies - Colorado Brand Guidelines; C.R.S. 24-37.5-101, et seq.
- Third-party resources - Bootstrap; CSS Validation Service; Namespaces (W3C)

7. Definitions

- Accessibility: The design of products, devices, services, or environments for people who experience disabilities.
- Cascading Style Sheet (CSS): A language that describes the style of an HTML document, enabling separation of presentation and content.
- Usability: The ease of use and learnability of a digital property or user interface.
- Usability Testing: The process of quantifying usability and measuring a user's experience to identify and drive improvements.
- User Centered Design: A design process in which the needs of end users are provided extensive attention at every stage.
- User Experience (UX): The overall experience of a person using a product.
- User Interface (UI): The method through which people can interact with input devices, applications, and websites.